

MGT 6081 - Derivative Securities
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Volatility Mispricing in Crude Oil:

A GARCH-Guided ATM Straddle Strategy

Submitted by

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1. Introduction

Commodities are the natural resources that drive the world's economy. With that said, they are among the most intriguing and complex asset classes to analyze because of the various factors, both economic and non-economic, that must be taken into consideration when working with these assets. Currently, amid the global geopolitical landscape, oil has been a major talking point in headlines and the broader macroeconomic landscape. As current QCF students, our interest in this space and the rigorous combination of the dynamic, ever-changing landscape were major motivations for this project. Additionally, the various factors that go into accurately analyzing this thought-provoking industry further piqued our interest, along with the mathematical complexity of derivatives. Specifically, we are developing an options trading strategy based on the difference between realized and implied volatility, which will be explained in further detail later. Since we know oil is a highly volatile asset, we will use a volatility strategy, such as a straddle, to capture the difference between realized and implied volatility, so we can do what any aspiring trader wants to do: make enormous returns.

2. Data

For our analysis, we considered from January 2007 to April 2026 to effectively represent different market conditions and political regimes to have the most robust results. Thankfully, this data was readily available on various classic financial databases where many financial professionals go to find this vital information. To collect this data, we used Yahoo Finance both for the CLF historical daily asset price data and the OVX index, which is a proxy for the 30-day implied volatility of oil prices. Additionally, we used the FRED database to get data for the 1-month risk-free rate, which is vital for the next part of our analysis. Once this data was effectively compiled, we had what we needed to be able to proceed.

To continue to build our trading strategy, one of the most important things that we need to do is price options. This was effectively done by using the Black-Scholes formula and the variety of data collected from Yahoo Finance and the FRED database so that we had the parameters necessary to price both ATM calls and puts. It is important to note that to price the ATM calls and puts, we used the OVX index to do so. However, since our entire trading strategy is based on the difference between realized volatility and implied volatility, we still needed to obtain the realized volatility. First, we took the historical CLF price data and converted these prices into log returns. Since these were daily returns, we needed to annualize the returns as well. Also, since OVX is a 30-day implied volatility, we must make sure that the realized volatility is treated in the same way so that we have an accurate analysis. Therefore, we need to have 30-day rolling windows to ensure that this is the case. Now, we calculated the mean volatility so that we could see when there were periods of high volatility and periods of lower volatility. This was a major reason why we wanted to have data from 2007 to 2026 to demonstrate and test our strategy in different market conditions and political regimes to see when the strategy performed best and if it still performed under difficult geopolitical and market combinations.

3. Technique

This section describes the methodological pipeline used to construct, calibrate, and back test the options strategy. The approach proceeds in four stages: option pricing, volatility modeling, signal generation, and portfolio construction with risk management.

3.1 Option Pricing: Black-Scholes Applied to Futures Prices

Crude oil options are written on futures contracts rather than on the spot price directly. We price options using a Black-Scholes style formula applied directly to the futures price, treating the WTI futures price F as the underlying in place of a spot price:

$$\begin{aligned}C &= e^{-rT} [F \cdot N(d_1) - K \cdot N(d_2)] \\P &= e^{-rT} [K \cdot N(-d_2) - F \cdot N(-d_1)] \\d_1 &= [\ln(F/K) + (r + \frac{1}{2}\sigma^2) T] / (\sigma\sqrt{T}), \quad d_2 = d_1 - \sigma\sqrt{T}\end{aligned}$$

where F is the WTI futures price, K is the at-the-money strike (rounded to the nearest \$0.50), r is the continuously compounded 1-month Treasury rate sourced from FRED (series DGS1MO), σ is the implied volatility drawn from the OVX index (expressed as a decimal), and $T = 30/365$ reflecting a 30-day holding horizon.

An at-the-money straddle is constructed each day by summing the call and put prices. The straddle is long both legs, giving a payoff equal to the absolute price move over the holding period minus the premium paid. This structure is appropriate for our hypothesis: if the market is overpricing volatility relative to what the GARCH model forecasts, the straddle will be cheap relative to the realized move that follows. The ATM call prices, put prices, and their sum are plotted below for the full historical sample.

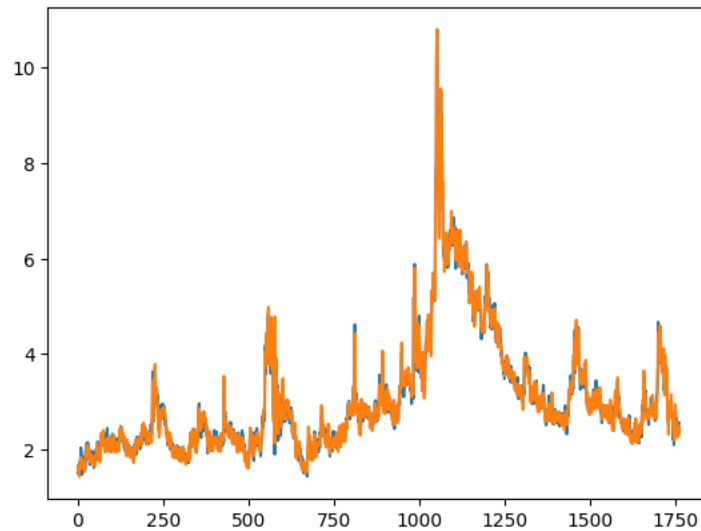


Figure 1. Daily ATM call and put prices computed via Black-Scholes applied to the futures price, using OVX as implied volatility input. The sharp spike around index ~1,000 corresponds to the COVID-19 volatility shock of April 2020.

Vega, the first-order sensitivity of the option price to a unit change in implied volatility, is also calculated daily and is used later as a position-sizing input. Higher vega implies greater volatility risk per dollar of premium, and positions are scaled down accordingly.

3.2 Volatility Modeling: GARCH(1,1) with Student-t Errors

The central question motivating the strategy is whether the OVX implied volatility index represents a fair forecast of future realized volatility, or whether it tends to be systematically elevated. To answer this, we fit a GARCH(1,1) model with Student-t distributed errors to the time series of WTI daily log returns, producing a competing volatility forecast that is purely statistical and contains no forward-looking market sentiment.

The GARCH(1,1)-t model is specified as:

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \sim tv$$

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

The Student-t specification is chosen over the standard Gaussian GARCH because WTI log returns exhibit well-documented fat tails: extreme moves like the April 2020 price collapse or recent geopolitical shocks occur far more frequently than a normal distribution implies. We begin with a plot of raw daily log returns to motivate this choice.

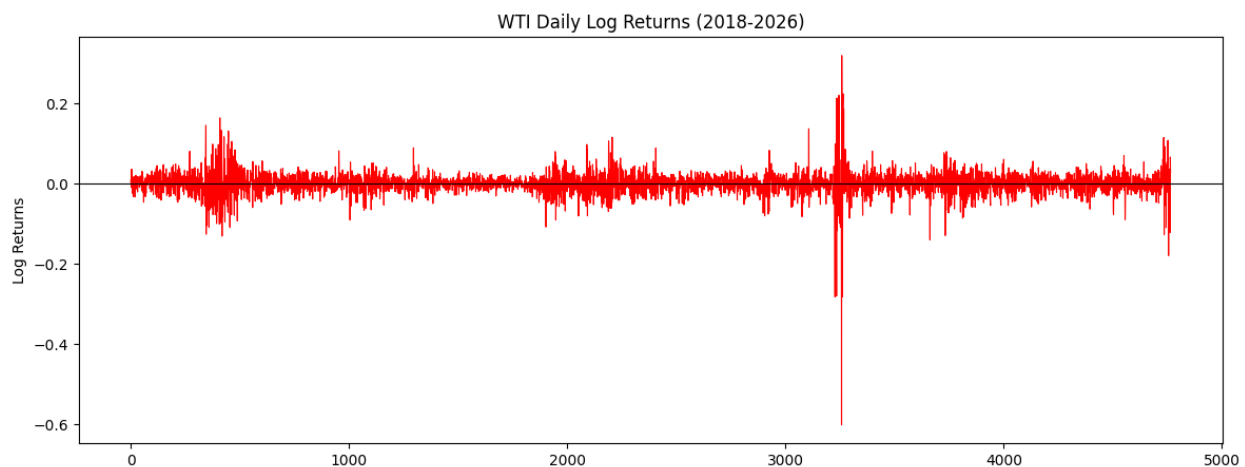


Figure 2. WTI crude oil daily log returns. The large negative spike around index ~3,300 corresponds to April 2020; note the fat-tailed distribution relative to a Gaussian baseline.

To prevent lookahead bias, the GARCH model is trained exclusively on the first 50% of observations (the in-sample period). The fitted model then generates rolling 30-day-ahead variance forecasts over the remaining out-of-sample period. Log returns are rescaled by a factor of 10 prior to fitting to address numerical convergence warnings, and the resulting volatility forecasts are rescaled back accordingly.

The 30-day rolling realized volatility and the GARCH conditional volatility are plotted individually and together below, providing a visual basis for comparing the two measures.

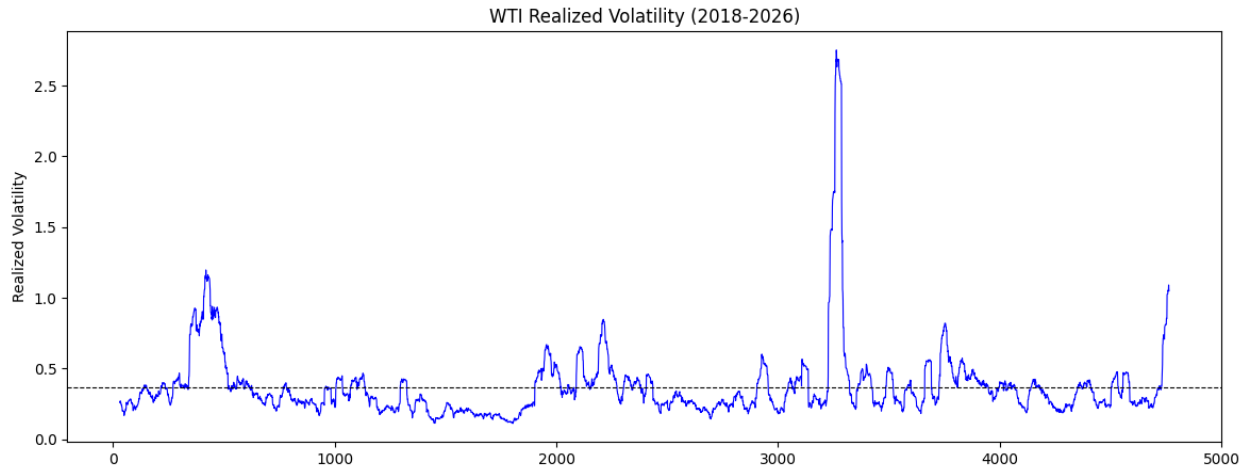


Figure 3. WTI 30-day rolling realized volatility (annualized). The dominant spike around index $\sim 3,300$ is the April 2020 COVID crash.

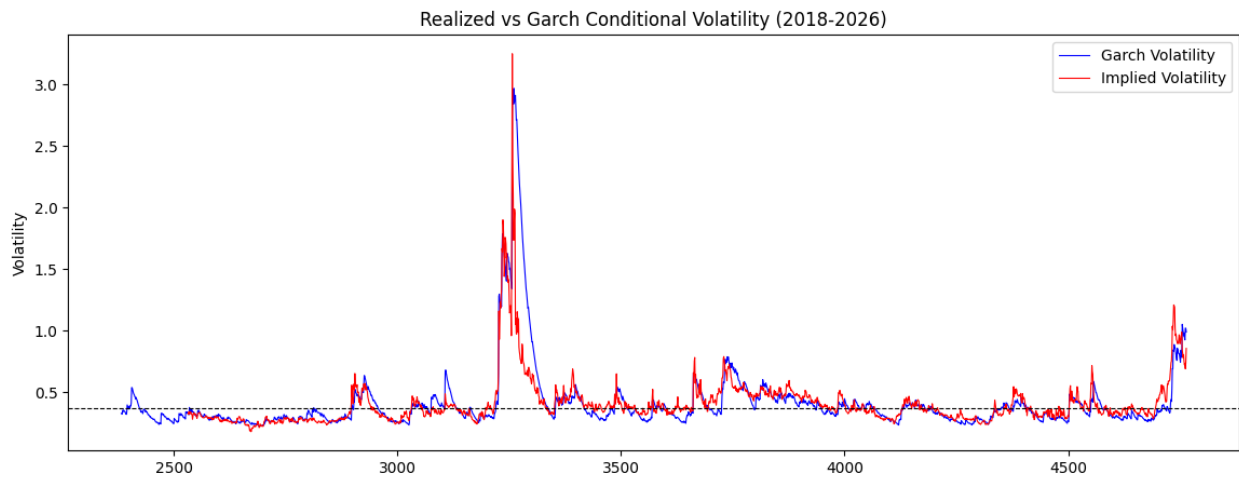


Figure 4. GARCH-forecasted volatility (blue) overlaid with OVX implied volatility (red) for the out-of-sample half of the data. OVX persistently sits above the GARCH estimate, particularly during stress episodes.

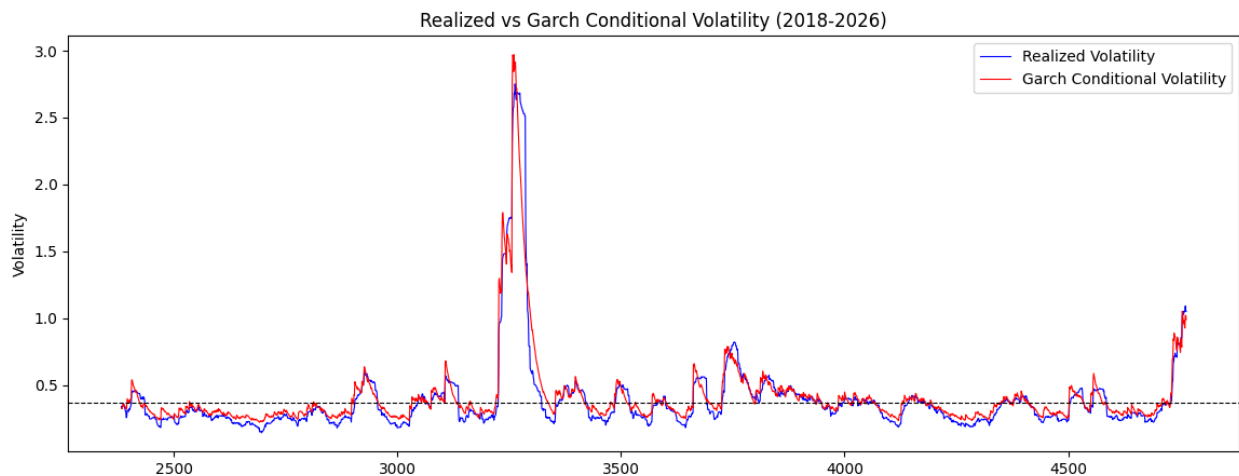


Figure 5. Realized volatility (blue) vs. GARCH conditional volatility (red). The two series track each other closely in calm regimes but diverge sharply during tail events.

3.3 Signal Generation: Z-Score of the Volatility Spread

With two volatility measures in hand, OVX implied volatility and the GARCH forecast, we define the volatility spread as their daily difference:

$$Spread_t = OVX_t - GARCH_t$$

A positive spread indicates the market is pricing in more volatility than the GARCH model suggests is warranted. To distinguish anomalous readings from ordinary variation, the spread is Z-scored over a rolling 150-day window:

$$Z_t = (Spread_t - \mu_{t^n}) / \sigma_{t^n}$$

The 150-day rolling window is chosen to provide a stable local baseline while remaining responsive to regime shifts. The raw spread and the resulting Z-score series are plotted below.

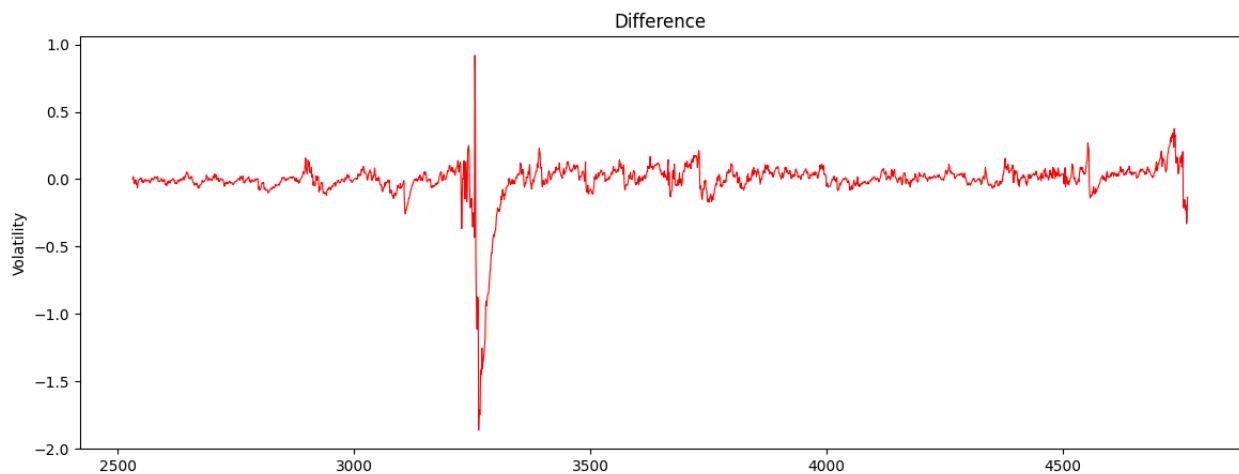


Figure 6. Raw daily spread (OVX – GARCH forecast). Positive values indicate market implied volatility exceeds the GARCH statistical forecast. The extreme negative dip corresponds to the April 2020 period when realized volatility briefly spiked above OVX.

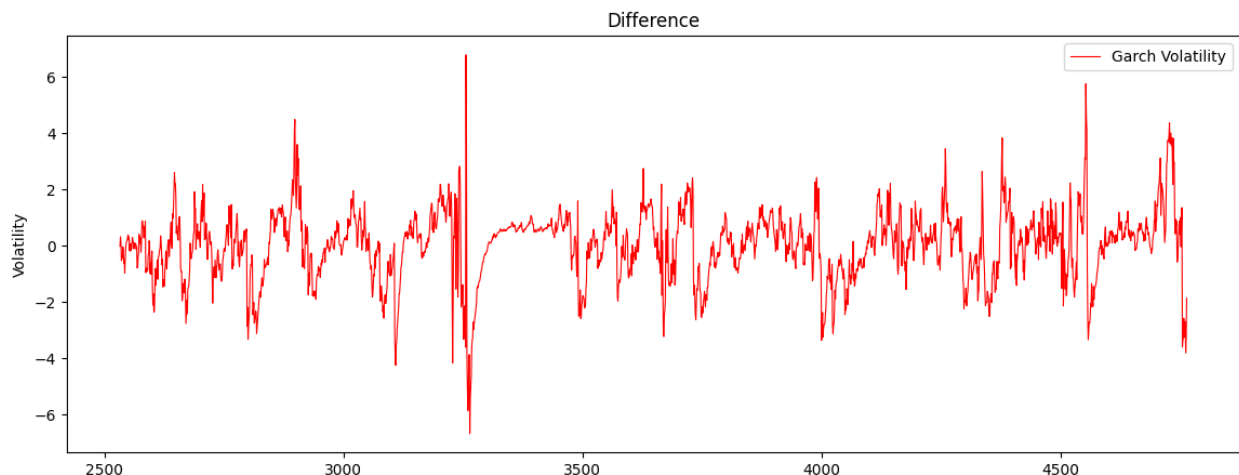


Figure 7. Rolling Z-score of the volatility spread. Values above the entry threshold of 1.25 trigger a long straddle signal. The series is bounded during calm regimes but spikes sharply during crisis episodes.

To validate the signal empirically before committing to a threshold, we conducted a systematic basket analysis: the historical sample was partitioned into 0.5-unit Z-score buckets spanning –3.0

to +3.5, and a long-straddle backtest was run independently within each bucket. The results are presented in Table 1 and Figure 8.

Table 1: Long Straddle Performance by Z-Score Bucket (Historical, 2007–2026)

Z-Score Range	Trades	Win Rate	Avg Return	Sharpe	Total Return
[-3.0, -2.5)	36	41.7%	-2.1%	-0.07	-6.98%
[-2.5, -2.0)	70	42.9%	+1.7%	+0.07	-1.44%
[-2.0, -1.5)	104	38.5%	-6.8%	-0.27	-29.58%
[-1.5, -1.0)	141	48.2%	+15.9%	+0.65	+83.37%
[-1.0, -0.5)	232	52.6%	+7.7%	+0.33	+39.43%
[-0.5, 0.0)	338	39.3%	-10.0%	-0.41	-79.28%
[0.0, 0.5)	480	44.6%	+5.9%	+0.20	+1.53%
[0.5, 1.0)	414	43.5%	+8.9%	+0.28	+0.61%
[1.0, 1.5)	189	59.8%	+67.3%	+1.23	+1,283%
[1.5, 2.0)	90	62.2%	+103.9%	+1.67	+627%
[2.0, 2.5)	40	57.5%	+70.8%	+1.40	+117%
[2.5, 3.0)	12	33.3%	+77.5%	+1.10	+27.83%
[3.0, 3.5)	8	25.0%	+15.8%	+0.30	+3.55%

Highlighted rows ($Z \geq 1.0$) indicate the primary entry zone.

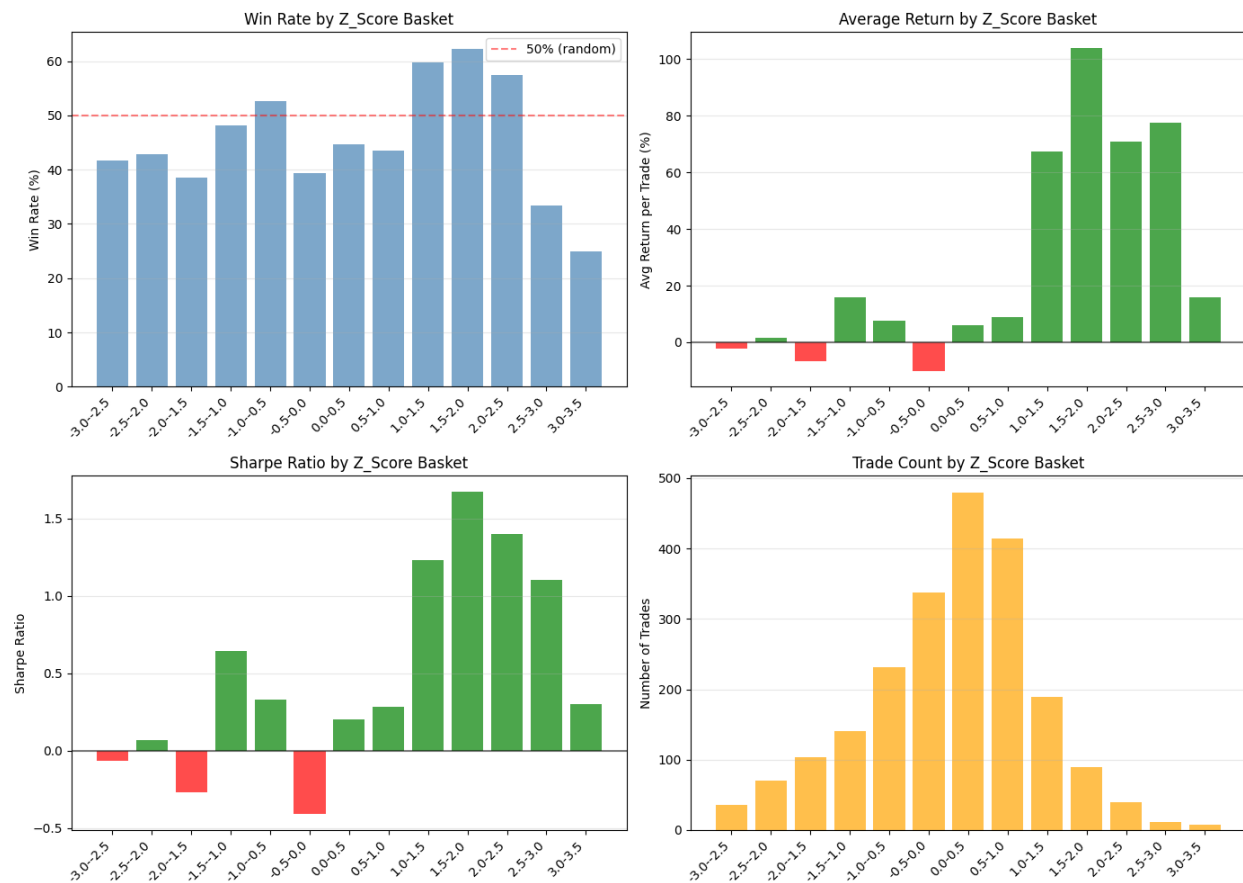


Figure 8. Four-panel summary of long straddle performance across Z-score baskets. Top-left: win rate; top-

right: average return per trade; bottom-left: Sharpe ratio; bottom-right: trade count. Green bars indicate positive returns; red bars indicate losses.

The results are clear. For Z-scores below 0, the strategy is largely unprofitable. The center of the distribution (Z near 0) generates no meaningful edge either. However, once the Z-score exceeds 1.0, both win rates and average returns improve dramatically. The sweet spot is the [1.0, 2.0) range, where 150-279 trades achieve win rates of 60-62% and Sharpe ratios of 1.2-1.7. Above Z = 2.5, the trade count drops sharply (12 and 8 observations respectively), making those estimates statistically unreliable.

Based on this analysis, the final entry threshold is set at $Z > 1.25$. No short-straddle signal is used: the short-vol direction was tested at $Z < -1.5$ and consistently lost money. The GARCH model appears to systematically underestimate realized volatility in calm regimes, making implied vol appear cheap when it is simply reflecting genuinely low uncertainty.

3.4 Portfolio Construction and Risk Management

When the Z-score exceeds 1.25, a long ATM straddle is entered and held for 30 calendar days. The trade payoff is the absolute difference between the terminal futures price and the original strike, minus the straddle premium paid:

$$PnL_t = |F_{t+30} - K| - \text{Straddle}_t$$

Position sizing uses a vega-scaled allocation rule. Because vega measures how much the straddle price changes with a unit move in implied volatility, a high-vega day implies greater volatility risk per dollar invested. Each trade's capital allocation is therefore scaled inversely with vega relative to a 120-day rolling median baseline, and clipped to stay within practical bounds:

$$\text{Alloc}_t = \text{Clip}[(1/50) \times (\text{Vega}_{\text{med}} / \text{Vega}_t), 1/90, 1/15]$$

This rule reduces exposure when options are expensive (high vega) and increases it modestly when they are cheap, while hard limits prevent any single trade from consuming more than 6.7% or less than 1.1% of portfolio capital. A maximum of 30 simultaneous open positions is enforced. The dynamic behavior of this allocation rule over the backtest period is shown below.

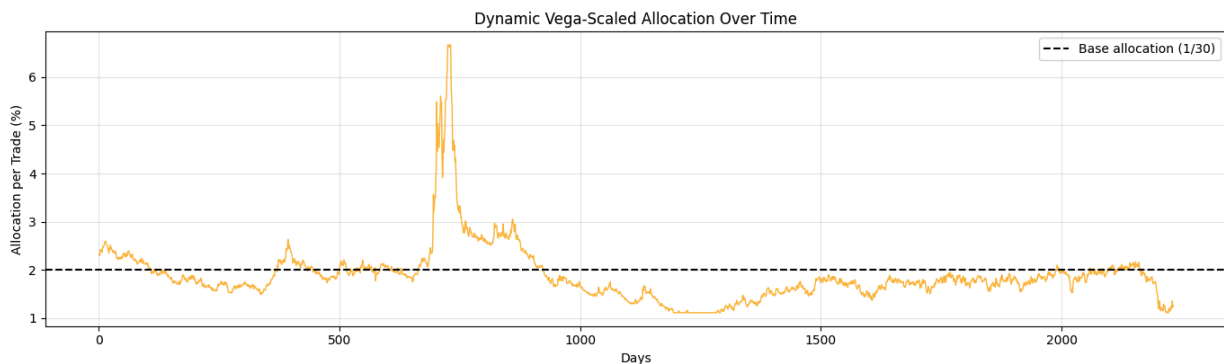


Figure 9. Vega-scaled allocation percentage per trade over the backtest period. The dashed line shows the base allocation of 1/50 (2%). Allocations spike upward during periods of low vega (cheap options) and compress during the COVID-era high-vega regime.

3.5 Forward Simulation: Heston Stochastic Volatility Model

To stress-test the strategy beyond the historical sample, we simulate future oil price paths using the Heston stochastic volatility model. Unlike Geometric Brownian Motion (GBM), the Heston

model allows variance to be stochastic and mean-reverting, which is more consistent with the well-documented volatility clustering and leverage effects observed in energy markets. The model is:

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW^1_t$$

$$dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW^2_t, \quad \text{Corr}(dW^1, dW^2) = \rho$$

where κ is the mean-reversion speed of variance, θ is the long-run variance level, ξ (ξ) is the volatility-of-volatility, and ρ is the correlation between the price and variance Brownian motions (negative for oil, reflecting the leverage effect). A separate mean-reverting process is co-simulated for the implied volatility (OVX proxy), which reverts toward the instantaneous Heston variance with its own noise term, ensuring the OVX and price series remain jointly consistent.

The simulation covers 1,008 trading days (~4 years) starting from the most recent observed WTI price of \$92.13. Three market regimes are parameterized in the code: a calm range-bound regime, a moderate bullish regime, and a stress selloff regime. The primary reported results use the calm regime, which is the most conservative and therefore the most critical test of strategy robustness. The other two regimes are available in the codebase for future analysis.

4. Results and Conclusions

4.1 Historical Backtest Performance

Applied to the full out-of-sample period (approximately 2014-2026 after the GARCH training split and rolling window burn-in), the vega-scaled long-straddle strategy starting from \$1,000,000 grows to \$14,888,631, a total return of 1,388.86%. Summary statistics are presented in Table 2.

Table 2: Historical Backtest Results — Vega-Scaled Long Straddle ($Z > 1.25$)

Metric	Value
Initial Capital	\$1,000,000
Final Equity	\$14,888,631
Total Return	1,388.86%
Number of Trades	247
Win Rate	59.1%
Average Return per Trade	85.56%
Average Allocation per Trade	2.04%
Raw Sharpe Ratio	1.49
Realized Sharpe Ratio	2.15
Raw Sortino Ratio	8.69
Realized Sortino Ratio	3.27
Maximum Drawdown	-20.82%

Out-of-sample results; zero transaction costs assumed.

The strategy achieves a win rate of 59.1%, a consistent ~9 percentage point edge over random selection. The realized Sharpe ratio of 2.15 is notably high. The Maximum Drawdown of -20.82% is manageable relative to the overall return, suggesting the vega-scaling rule provides meaningful downside protection. The full equity curve, cumulative return series, and dynamic allocation are shown below.

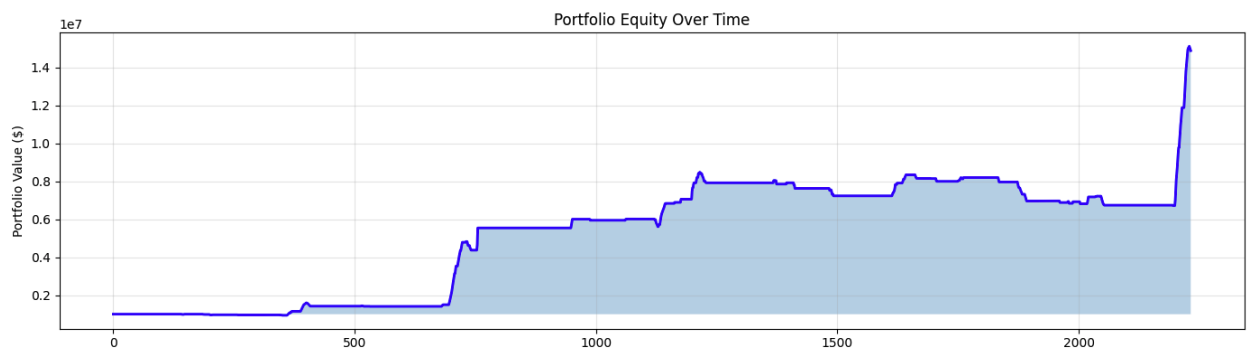


Figure 10. Portfolio equity curve from the historical backtest. Two sharp upward steps are clearly visible: one around day ~1,000 (COVID-19 shock, 2020) and one at the far right (2025–2026 macro volatility episode). The

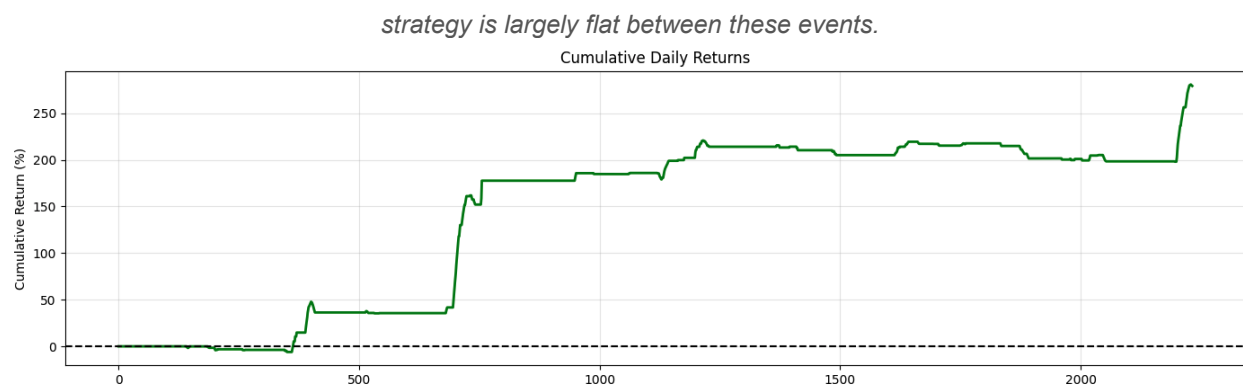


Figure 11. Cumulative daily returns (%) over the backtest period. The flat profile between volatility events highlights the strategy's dependence on high-regime episodes.

Examining the equity curve, two distinct performance bursts account for the majority of the total return. The first corresponds to the COVID-19 volatility shock of March-April 2020, when WTI crude famously turned negative and OVX surged to historic highs. The second, more recent burst corresponds to the 2025-2026 period of elevated macro and geopolitical uncertainty, during which OVX reached 0.86 on April 21, 2026. Between these two events, the portfolio grows slowly, spending extended periods near flat.

4.2 Forward Simulation Performance (Heston, Calm Regime)

Under the Heston calm regime simulation (1,008 days, starting price \$92.13), the strategy generates 69 trades. Full simulation statistics are reported in Table 3.

Table 3: Forward Simulation Results — Heston Calm Regime

Metric	Value
Number of Trades	69
Win Rate	66.7%
Mean Return per Trade	57.7%
Annualized Return	484.6%
Annualized Std. Deviation	319.5%
Trade Sharpe Ratio	1.52
Realized Sharpe Ratio	1.95
Total Uncompounded Return	3,980.97%

Calm range-bound regime; seed = 1; 4-year simulation horizon.

The strategy performs strongly in simulation, with a higher win rate (66.7% vs. 59.1%) than in the historical backtest. This is partly an artifact of the simulation parameterization: the calm Heston regime produces stable, predictable Z-score behavior without the extreme fat-tail events that dominate the historical results. Sharpe ratios of 1.5-2.0 are consistent with those observed historically.

It is worth noting that the Maximum Drawdown as computed on trade-level uncompounded returns outputs -100% in this simulation, which is a known numerical artifact of the metric rather than a

reflection of total capital loss. A daily equity-curve MDD, as computed in the historical backtest (-20.82%), would be the appropriate measure in a live implementation, and future iterations should report it consistently.

4.3 Limitations and Shortcomings

While the results are promising, several important limitations must be acknowledged before drawing strong conclusions about real-world viability.

- **Return concentration.** The overwhelming majority of the historical profit is driven by two volatility events: the COVID-19 crisis of 2020 and the 2025–2026 macro volatility episode. In the long stretches between these events, roughly 2013-2019 and 2021-2024, the strategy produces little to no return. A live implementation would require significant patience and capital discipline through extended flat periods, during which the strategy continuously pays option premium without recouping it.
- **Transaction costs.** All backtests assume zero transaction costs. In practice, ATM crude oil options carry meaningful bid-ask spreads, and straddles involve two legs, doubling the spread cost. Realistic costs of 3-5% of premium could substantially reduce or eliminate the edge in lower-return regimes, and would make the frequent low-Z-score entry attempts particularly costly.
- **Lookahead bias risk.** While the GARCH model is fitted only on in-sample data, the 150-day rolling Z-score normalization window uses observations that would have been contemporaneously available in real time. The careful training/test split mitigates the most serious data snooping concerns, but the threshold of $Z > 1.25$ was selected after observing the full basket analysis, which itself uses all out-of-sample data. In a true walk-forward setting, the threshold selection would need to be embedded inside the rolling window.
- **OVX availability and representativeness.** The OVX index did not exist prior to May 2007, limiting historical depth. Furthermore, the OVX measures the 30-day implied volatility of crude oil options traded on the CBOE, and may not perfectly reflect the implied volatility embedded in the specific front-month futures contracts being priced.
- **Short-vol leg abandoned.** The hypothesis that $Z < -1.5$ should predict short-straddle profitability was tested and found to be empirically invalid. The GARCH model appears to systematically underestimate volatility in calm regimes, making implied vol appear cheap when it is simply reflecting genuinely low uncertainty. As a result, the strategy is directional in its vol signal and it only benefits from overstated implied volatility, not understated.
- **Single-asset scope.** The strategy is tested only on WTI crude oil. Whether the GARCH-implied-vol spread signal generalizes to other commodities, equity indices, or rate products is unknown and represents a significant open question.

4.4 Avenues for Further Research

Several natural extensions could address the limitations above and potentially improve strategy robustness:

- **Stochastic interest rates.** Incorporating Ho-Lee, Vasicek, or CIR interest rate dynamics into the simulation framework would be particularly relevant for longer-dated options, where the risk-free rate assumption has a greater impact on option pricing.

- Richer volatility models. EGARCH, GJR-GARCH, or Realized GARCH using high-frequency intraday data may produce sharper forecasts and a more informative spread signal, especially in low-volatility regimes where the current GARCH model appears to underforecast.
- Machine learning signal enhancement. The original project plan included classification-based ML models using volatility and return features as inputs. Integrating a classifier for example, predicting whether the realized move will exceed the straddle premium could complement the Z-score threshold and improve trade selectivity, particularly in borderline signal regimes.

4.5 Concluding Remarks

This project demonstrates that the spread between OVX-implied volatility and GARCH-forecasted volatility contains statistically meaningful information about future straddle profitability in crude oil markets. When the Z-score of this spread is sufficiently elevated, indicating that market fear is highly relative to recent statistical volatility, the long ATM straddle strategy achieves a consistent ~60% win rate and Sharpe ratios above 1.5, both in historical backtesting across nearly two decades of data and under Heston forward simulation.

These results are consistent with a well-documented phenomenon in commodity options markets: implied volatility carries a positive risk premium over realized volatility on average, but this premium is not uniformly distributed. It concentrates during genuine episodes of uncertainty and fear. The GARCH model, by providing a purely backward-looking volatility estimate, serves as a useful anchor against which to measure when the market's forward-looking fear appears anomalously elevated, and therefore when straddles are attractively priced relative to the move likely to follow.

The primary caveat is that the strategy's positive performance is heavily driven by a small number of high-volatility events. This is both a feature and a limitation: it is precisely during these episodes that straddles produce outsized returns, but the strategy spends long periods generating little return while paying option premium. A practitioner deploying this strategy would need to account for this carry cost and would benefit from the extensions described in Section 4.4, particularly the addition of transaction costs, a true walk-forward threshold selection procedure, and expanded multi-regime simulation coverage.

Overall, the project successfully implements a coherent pipeline from raw data acquisition through volatility modeling, signal generation, portfolio construction, and forward simulation, producing results that are analytically grounded and meaningfully positive under the conditions tested, while remaining transparent about the boundaries of those conditions.